Group and solo flight: comparisons

Group time against solo flight, raw, on the same route, by factor and around the group, on open flights

Matteo Di Sante

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Abstract

We compare paraglider group time with solo flight on open flights, four ways: raw; against solo flights on the same route, season, hour and wing; by how much each of those factors accounts for the raw gap; and against the same flight before and after its group time. Raw, group time progresses straighter than solo flight: $H_{\text{loc}}(1000\text{ s})$ is 0.915 in G and 0.897 in S. On the same route, season, hour and wing the gap is $+0.008$ [-0.003, $+0.025$], a band that includes 0; so do the bands of 9 of the 9 quantities. Taking the factors apart, the route carries the largest part of the difference of time in climb, glide speed and height lost per km of glide. Around the group, the drift-free effect g of $H_{\text{loc}}(1000\text{ s})$ is $+0.023$ [$+0.018$, $+0.029$]. Time in search, climb rate, glide speed and height lost per km of glide have opposite signs in the matched and in the before-and-after comparison.

1 Aim

Does a paraglider move differently in a group than alone? This report compares group time with solo flight on open flights in four ways, each on the same populations and measures.

1. **Raw** (Section 5): S, A and G as they are.
2. **Same route** (Section 6): pieces of group time against solo segments on the same route, in the same season and hour, with the same wing.
3. **Decomposition** (Section 7): how much of the raw gap the route, the season and hour, and the wing each account for.
4. **Before, during, after** (Section 8): the same flight in the windows around its group time.

Section 2 defines the flights, the populations and the measures; Sections 3 and 4 say which flights are open and why. Every table and figure gives the flights or pieces it rests on, its dataset and its estimator. Appendices give the formulas of each tool.

2 Flights, populations and measures

Which flights, which stretches of flight, and how are they measured?

Declared open flights. On the FFVL catalogue the pilot declares the type of the flight (Table 1). A distance, free or with up to three turn points, is open; a triangle, a quadrilateral or an out-and-return is closed; a hike and fly is neither. We keep the declared open flights: 66 845 of the 154 598 paraglider flights that depart and are logged every 10 s or finer (87 484 are closed, 269 neither).

Groups. The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. The groups are those of every flight of the archive, whoever the companions.

Populations. After its departure, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo. Three populations:

label	meaning	class	flights
Dist 3 pts	distance, 3 turn points	open	40 094
Dist 2 pts	distance, 2 turn points	open	25 399
Dist libre	free distance	open	694
Dist 1 pt	distance, 1 turn point	open	658
triangle	triangle	closed	51 181
triangle FAI	FAI triangle	closed	34 979
Quadrilatère	quadrilateral	closed	703
Aller-Retour	out and return	closed	617
Marche et Vol bouclé	hike and fly, closed	closed	3
Marche et Vol FAI	hike and fly, FAI triangle	closed	1
Marche et Vol	hike and fly	neither	269

Table 1: The labels of the FFVL catalogue, their class and their flights (paragliders that depart, logged every 10 s or finer).

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure;
- G, *group time*: the stretches in a group of the flights that entered one at least once;
- A, *the same flights alone*: the solo stretches of those same flights, with the same pilots, gliders, loggers and days as G.

Measure. A *piece* is a stretch within one segment of the track, on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta_f^2}(\tau) = \frac{1}{N_f(\tau)} \sum_t \left\| \mathbf{r}(t+\tau) - \mathbf{r}(t) \right\|^2, \tag{1}$$

with t and $t+\tau$ in the same piece. The ensemble $\mathrm{MSD}(\tau)$ is its mean over the flights, one weight per flight. $H_{\mathrm{loc}}(\tau)$ is half the slope of $\log \mathrm{MSD}$ against $\log \tau$ over the lags within ± 0.25 decades of τ: 1 for a straight line, $\frac{1}{2}$ for a random walk. H is half the slope over all the lags of a curve, from 10 s to its end. Bands are 5–95% of 1 000 bootstrap draws that resample the dependency blocks of the grouping (Appendix H).

Datasets. A lag τ needs pieces of at least $\tau/0.8$ s, so that the origins of every piece cover the lag in the same share of it. A dataset to the lag T keeps, in each population, its pieces of at least $T/0.8$ s and measures them at every lag up to T: the population does not change along a curve (Appendix C). Three datasets:

- *1000*: $T = 1\,000$ s, pieces of at least 1 250 s;
- *2250*: $T = 2\,250$ s, pieces of at least 2 812.5 s;
- *longest*: $T = 11\,450$ s, the longest lag at which each population keeps 100 flights, pieces of at least 14 312.5 s.

The quoted values are $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000 and $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on dataset 2250, the lag of the grid nearest to 2000 s. The window of $H_{\mathrm{loc}}(1840\ \mathrm{s})$, ± 0.25 decades, is cut at 2250 s: it uses the lags from 1 035 to 2250 s only.

Phase statistics. Flights logged at 1 Hz carry the phases of Vilpellet's method: climbs, searches and glides. Phases exist on 1-Hz flights only, so the phase statistics rest on fewer flights than the MSD (27 665 in S, 6 397 in A and 7 896 in G against 43 977, 8 991 and 11 077 on dataset 1000). A run belongs to the unit that holds its middle. Seven statistics, each a ratio n/d of two terms summed over the runs of a unit (Table 2); the phase time T_u of a unit u is the sum of the durations of its runs.

statistic	numerator n	denominator d	runs counted	r_u, q_u
time in climb (%)	time in climbs	phase time	climbs	n_u/T_u, constant
time in search (%)	time in searches	phase time	searches, found and vain	n_u/T_u, constant
searches ending in a climb (%)	searches followed by a climb	searches	searches	n_u/T_u, d_u/T_u
climbs per hour	climbs	phase time (h)	climbs	n_u/T_u, constant
climb rate (m/s)	height gained in climbs	time in climbs	climbs	n_u/T_u, d_u/T_u
glide speed (km/h)	straight-line distance of glides	time in glides	glides	n_u/T_u, d_u/T_u
height lost per km of glide (m/km)	height lost in glides	distance of glides	glides	n_u/T_u, d_u/T_u

Table 2: The seven phase statistics. Unit u of the average: the flight in Section 5 (its runs on its pieces of dataset 1000 in the population), the piece or control segment in Sections 6 and 7, the window in Section 8.

Estimator: average first, ratio after. With w_u the weight of a unit (1 for the value, the bootstrap count for a draw),

$$r_u = \frac{n_u}{T_u}, \qquad q_u = \frac{d_u}{T_u}, \qquad \text{stat} = \frac{\sum_u w_u\, r_u}{\sum_u w_u\, q_u}. \tag{2}$$

For the two time shares and the climbs per hour q_u is constant, and the statistic is the mean of the unit's share or rate. For the other four it does not divide by the few events of one unit: a unit with one search gives 0% or 100%. This is the estimator of the MSD, one weight per unit. The ratio of sums, $\sum n_u / \sum d_u$, weights a unit by its phase time (Appendix D).

Survival of the phase durations. For climbs, searches that end in a climb and glides, the survival function $P(\tau_{\text{phase}} > \tau)$ of the number of fixes of a run (1 Hz, so seconds), with the duration trimmed as in the note on phases. Every run weighs 1; its band is pointwise 5–95% of the draws of the unit's bootstrap (Appendix E).

3 Group, solo and alone on the declared open flights

How do the three populations compare as they are, with no stratification?

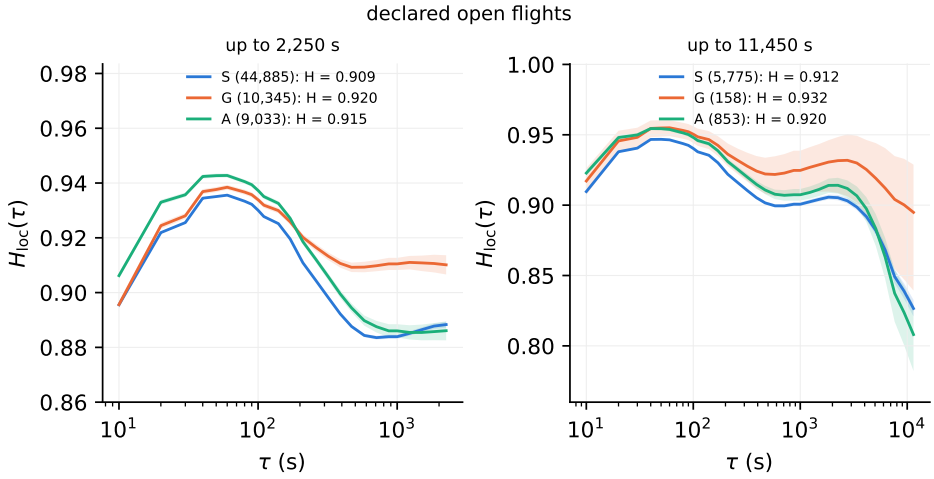


Figure 1: Declared open flights. $H_{\text{loc}}(\tau)$ of S, G and A for the datasets up to 2 250 s (left) and 11 450 s (right), each on the pieces of at least $T/0.8$ s, with their flights and the fitted H in the legend; 5–95% bands. Flights of each curve in Table 3.

dataset	S	G	A
lags up to 2 250 s	44 885	10 345	9 033
lags up to 11 450 s	5 775	158	853

Table 3: Flights of each dataset of Figure 1, declared open flights.

On the dataset up to 2 250 s group time is the straightest (Figure 1): $H_{\text{loc}}(1000\text{ s})$ is 0.910 in G, against 0.884 in S and 0.886 in A, and the fitted H is 0.920, 0.909 and 0.915. On the dataset up to 11 450 s the mean squared displacement of S and A bends down at lags of hours: at 11 450 s H_{loc} is 0.827 in S and 0.808 in A. That of G stays at 0.895 [0.839, 0.929], on 158 flights only; the fitted H is 0.932, 0.912 and 0.920.

What we keep. Raw, group time progresses straighter than solo flights and than the same flights alone, and at lags of hours the motion of solo flights and of the same flights alone bends away from a power law.

4 How open the declared open flights are

Why does the mean squared displacement bend down at long lags?

Opening. The *opening* of a flight is the distance from its takeoff to its landing, its first and last fix, over the path flown: 1 for a straight line, near 0 for a flight that comes back to its takeoff.

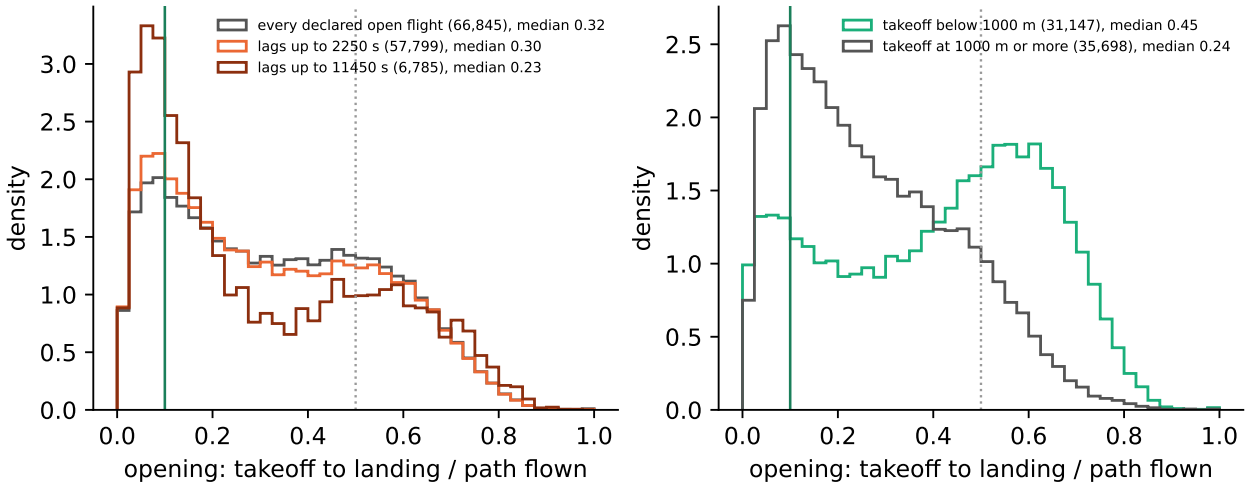


Figure 2: Opening of the declared open flights, takeoff to landing over the path flown. Left: every declared open flight and the flights of the datasets of Figure 1. Right: takeoffs below and above 1 000 m. Solid, the threshold 0.1; dotted, the stricter 0.5 that we do not use.

	flights				median opening			
dataset	all	S	G	A	all	S	G	A
every declared open flight	66 845				0.32			
lags up to 2 250 s	57 799	44 885	10 345	9 033	0.30	0.31	0.28	0.23
lags up to 11 450 s	6 785	5 775	158	853	0.23	0.23	0.29	0.19

Table 4: Flights and median opening of the datasets of Figure 1, by population; a flight of G and A counts in each population where it has a piece long enough.

The long lags rest on folded flights. The declared open flights are far from straight lines: their median opening is 0.32 (Figure 2, Table 4). The longer the lag, the more folded the flights it rests on: the median opening is 0.30 on the dataset up to 2 250 s and 0.23 up to 11 450 s, and the share of flights with an opening below 0.1 grows from 16% to 18% and 26%. A flight that comes back near its takeoff moves away and then back, and its displacement stops growing as the lag nears its duration: the bend of Section 3 is the shape of the flights. In the longest dataset the flights of G are more open than those of S and A (median 0.29 against 0.23 and 0.19), so part of the straighter progress of G at long lags is their shape.

The threshold. H has a meaning where the mean squared displacement is a power law, a straight line in log–log, and we want it on flights that are open. A threshold on the opening drops the declared open flights that are not. A strict one, 0.5, keeps only straight flights and raises H by construction; it also keeps 42% of the flights that take off below 1 000 m but only 13% of those above, where relief and airspace do not let a flight go straight (Figure 2, right), so it would select the region as well as the shape. We set the threshold at 0.1: the lowest that drops the flights that come back near their takeoff, it keeps nearly the same share of the two, 88% and 80%. It keeps 55 879 flights, 84% of the declared open ones.

What we keep. From here on an *open flight* is a declared open flight with an opening above 0.1.

5 Point 1: raw

How do S, A and G compare as they are, on the open flights, with no stratification?

The MSD. On dataset 1000, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.915 [0.913, 0.918] in G, 0.897 [0.897, 0.898] in S and 0.894 [0.892, 0.897] in A (Table 5, Figure 3): the gap G − S is +0.018, and the bands of G and S do not overlap. The three curves agree at short lags; at 1000 s G stays high while S and A have fallen. The same flights alone follow S. On dataset 2250, $H_{\mathrm{loc}}(1840\ \mathrm{s})$ is 0.923 in G and 0.906 in S, a gap of +0.017. On the longest dataset the three curves rise and G keeps 115 flights only: $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.936 [0.919, 0.950] in G and 0.912 in S (Figure 14, Appendix L).

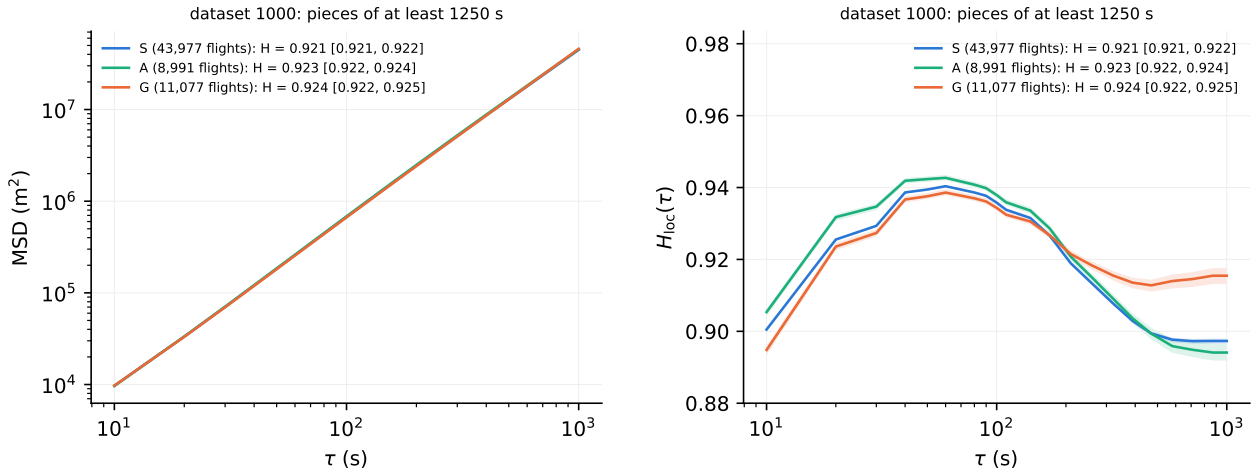


Figure 3: Open flights, dataset 1000 (pieces of at least 1 250 s, lags to 1 000 s). Left: ensemble MSD(τ), one weight per flight; right: $H_{\text{loc}}(\tau)$; S (43 977 flights), A (8 991) and G (11 077), fitted H in the legend; 5–95% bands of the draws of dependency blocks.

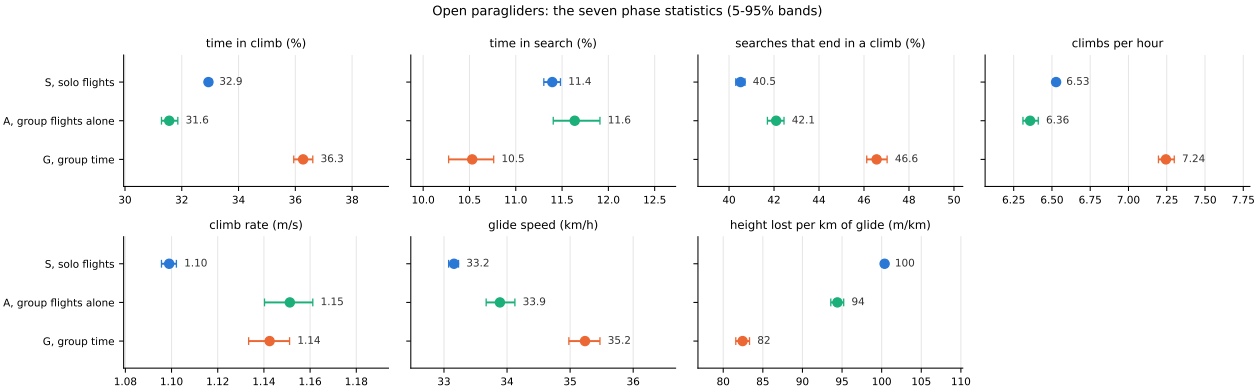


Figure 4: Open flights: the seven phase statistics of S (27 665 flights), A (6 397) and G (7 896), on the pieces of dataset 1000; average first, ratio after, unit the flight; 5–95% bands.

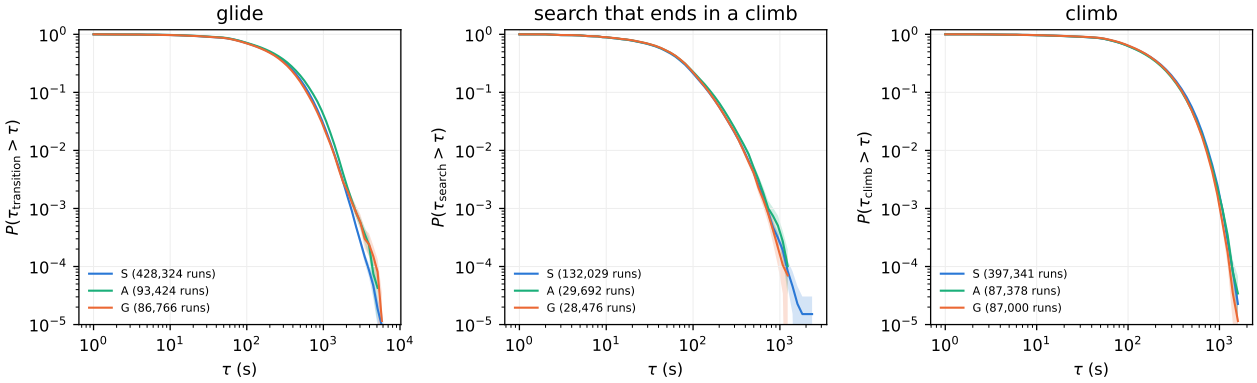


Figure 5: Open flights: survival $P(\tau_{\text{phase}} > \tau)$ of the duration of glides, of searches that end in a climb and of climbs, on the pieces of dataset 1000, every run weighing 1; runs in the legend; pointwise 5–95% bands.

dataset		flights	pieces	fitted H	$H_{\mathrm{loc}}(1000\ \mathrm{s})$	$H_{\mathrm{loc}}(1840\ \mathrm{s})$
to 1 000 s	S	43 977	46 239	0.921 [0.921, 0.922]	0.897 [0.897, 0.898]	–
	A	8 991	13 464	0.923 [0.922, 0.924]	0.894 [0.892, 0.897]	–
	G	11 077	14 401	0.924 [0.922, 0.925]	0.915 [0.913, 0.918]	–
to 2 250 s	S	36 975	37 865	0.916 [0.915, 0.916]	0.898 [0.897, 0.899]	0.906 [0.905, 0.907]
	A	6 966	8 779	0.920 [0.919, 0.921]	0.897 [0.895, 0.900]	0.902 [0.899, 0.905]
	G	8 405	10 134	0.923 [0.922, 0.925]	0.918 [0.915, 0.920]	0.923 [0.920, 0.926]
to 11 450 s	S	4 309	4 311	0.923 [0.922, 0.924]	0.912 [0.911, 0.914]	0.922 [0.920, 0.923]
	A	602	602	0.929 [0.926, 0.934]	0.917 [0.912, 0.922]	0.926 [0.921, 0.932]
	G	115	115	0.943 [0.928, 0.954]	0.936 [0.919, 0.950]	0.948 [0.931, 0.962]

Table 5: Open flights, raw: flights and pieces of each population and dataset, fitted H (10 s to the end of the lags), $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and $H_{\mathrm{loc}}(1840\ \mathrm{s})$, with 5–95% bands. $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000 is measured at the lag 1000 s; $H_{\mathrm{loc}}(1840\ \mathrm{s})$ needs a dataset of at least 2250 s, and its window is cut at 2250 s. Ensemble mean over flights, one weight per flight.

The phases. G spends more time in climb than S (36.3% against 32.9%), finds a climb after a search more often (46.6% against 40.5%), makes more climbs per hour (7.24 against 6.53), glides faster (35.2 against 33.2 km/h) and loses less height per km of glide (82 against 100 m/km); its time in search is lower (10.5% against 11.4%). The climb rate is 1.14 against 1.10 m/s. For all statistics but the climb rate A lies nearer to S than to G. The survival functions of the three populations (Figure 5) overlap for climbs and for searches; for glides the tail of S falls below those of A and G at the longest durations, where few runs remain.

The estimator matters. The ratio of sums weights a flight by its phase time. It gives 34.2% of time in climb in S against 32.9% for the average first, and 34.2 against 33.2 km/h of glide speed (Appendix D, Table 11).

What we keep. Raw, on the open flights, group time progresses straighter than solo flight ($H_{\mathrm{loc}}(1000\ \mathrm{s})$: 0.915 against 0.897) and gains more height per unit of time: more time in climb, more climbs per hour, a glide that is faster and loses less height. The same flights alone stay close to S. This gap mixes the route, the season, the hour, the wing and the flights themselves, which the next two sections take apart.

6 Point 2: the same route, season, hour and wing

Does group time still differ from solo flight when the solo side flies the same route, in the same season and hour, with the same wing?

Controls. For each piece of G (a stretch in a group, at least 1 250 s) the controls are the solo segments of the pool: the 4 518 open flights that never had a contact of any length, on another date than the piece. A segment matches when at least 0.8 of the points of the piece, one every 30 s, have the pool track within 500 m, the pool flight flew them in the same direction, and the segment lasts at least as long as the dataset needs (Appendix F). The match table holds 96 462 matches; 6 018 pieces of G and 3 851 pieces of A have at least one. A caliper then keeps the matches of the same season (month), the same hour and the same wing.

	G, dataset to 1000 s			G, dataset to 2250 s			A, all	
caliper	beg.	exp.	all	beg.	exp.	all	1000	2250
pieces without a condition	3 940	10 002	14 401	2 679	7 132	10 134	13 464	8 779
$*$ month ±1, hour ±1, same class	783	973	1 763	311	395	708	994	277
month ±1, hour ±1, same level	824	1 511	2 335	327	656	983	1 299	376
month ±2, hour ±2, same class	1 132	1 585	2 730	513	725	1 241	1 616	520
month ±2, hour ±2, same level	1 181	2 251	3 432	537	1 095	1 632	1 998	677

Table 6: Pieces of G (dataset 1000 and 2250) and of A with at least one control, by caliper, by level (beg. beginners, exp. experienced, wings A and B against C, D and CCC); $*$ the chosen caliper. Month and hour are those of the middle of the piece and of the segment.

The caliper. The calipers are tried in the order of Table 6, and the chosen one is the first with at least 200 pieces of G per level on dataset 2250: month ±1, hour ±1, the same wing class (311 pieces for beginners, 395 for experienced). It leaves 1 763 pieces of G on dataset 1000, from 1 648 flights and 1 040 pilots, and 994 of A. Isolation (nobody within 10 km) is off in the main result.

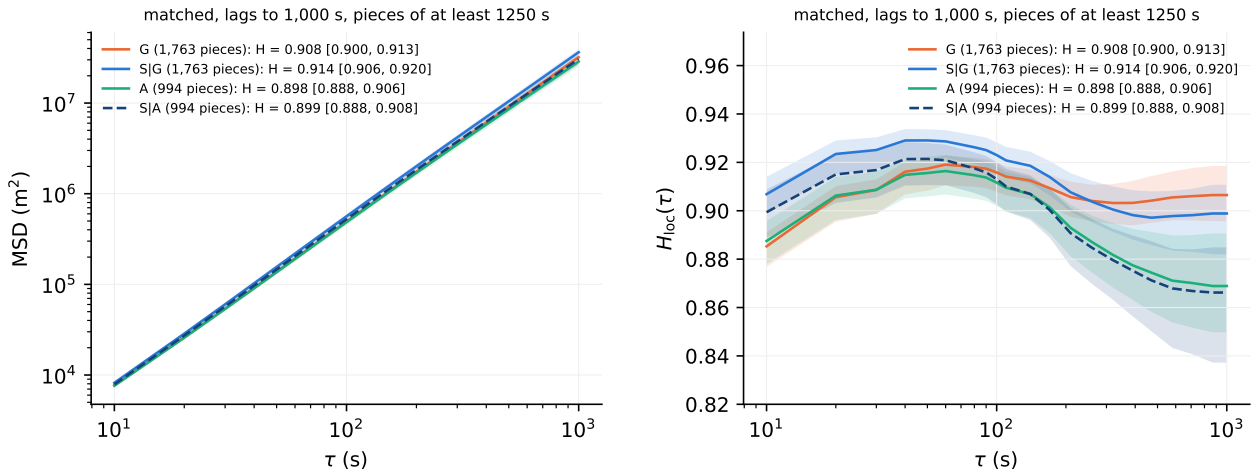


Figure 6: Same route, season, hour and wing (caliper month ±1, hour ±1, same class), dataset 1000 (pieces of at least 1 250 s). Left: ensemble MSD(τ); right: $H_{\mathrm{loc}}(\tau)$. G (1 763 pieces), its controls S | G (3 444 segments), A (994 pieces) and S | A (1 827); each side averaged over units reweighted to the piece (Appendix G); 5–95% bands of the draws of pilots and blocks.

	pieces	fitted H, 10 to 1000 s	H_{loc}(1000 s)	H_{loc}(1840 s)
G	1 763	0.908 [0.900, 0.913]	0.906 [0.896, 0.919]	0.893 [0.868, 0.916]
S | G	3 444	0.914 [0.906, 0.920]	0.899 [0.882, 0.911]	0.875 [0.840, 0.900]
A	994	0.898 [0.888, 0.906]	0.869 [0.850, 0.891]	0.862 [0.811, 0.909]
S | A	1 827	0.899 [0.888, 0.908]	0.866 [0.837, 0.885]	0.843 [0.778, 0.897]
G − S | G		−0.005 [−0.012, −0.001]	+0.008 [−0.003, +0.025]	+0.018 [−0.002, +0.048]
A − S | A		−0.002 [−0.010, +0.008]	+0.003 [−0.014, +0.032]	+0.019 [−0.027, +0.070]
difference		−0.004 [−0.015, +0.006]	+0.005 [−0.023, +0.027]	−0.001 [−0.055, +0.051]

Table 7: Same route, season, hour and wing, level all: pieces (or control segments), fitted H (dataset 1000, 10 to 1000 s), H_{loc}(1000 s) (dataset 1000) and H_{loc}(1840 s) (dataset 2250, 708 pieces of G); differences of the sides with 5–95% bands. Ensemble mean over units reweighted to the piece.

The MSD. Against its controls, G has H_{loc}(1000 s) of 0.906 and S | G of 0.899: Δ(G − S | G) is +0.008 [−0.003, +0.025] (Table 7, Figure 6), against the raw gap +0.018 of Section 5. On dataset 2250, H_{loc}(1840 s) gives +0.018 [−0.002, +0.048]. The two bands include 0. The fitted H is lower in G than in its controls (−0.005 [−0.012, −0.001]): its H_{loc} is lower than that of its controls at short lags and higher at the longest lags of dataset 1000. The band of Δ(G − S | G) includes 0 for 9 of the 9 quantities of Figure 7.

Selection. A against S | A flies the same route alone: same pilots, gliders and days as G, no group. Δ(A − S | A) measures selection. Its H_{loc}(1000 s) is +0.003 [−0.014, +0.032] and the band includes 0 for 9 of the 9 quantities; the difference (G − S | G) − (A − S | A) is +0.005 [−0.023, +0.027] at H_{loc}(1000 s) and includes 0 for 9 of them.

The phases. Time in climb is 29.2% in G and 26.3% in its controls (Δ is +2.9 [−0.1, +5.6]), climbs per hour 7.63 against 6.98, glide speed 29.5 against 30.9 km/h (−1.4 [−2.8, +0.4]) and the climb rate Δ is −0.054 [−0.162, +0.037] m/s. The survival functions of the three phases (Figure 8) overlap within their bands.

Robustness. Over the 9 lines of Figure 9, ΔH_{loc}(1000 s) stays between +0.007 and +0.012, with a band that includes 0 in 7 of them, and ΔH_{loc}(1840 s) between +0.016 and +0.031, with a band that includes 0 in 6.

What we keep. On the same route, season, hour and wing the gap of H_{loc}(1000 s) falls from +0.018 (raw) to +0.008 [−0.003, +0.025], and the bands of H_{loc}(1000 s), H_{loc}(1840 s) and of the seven statistics include 0 for 9 of the 9. The selection term Δ(A − S | A) is +0.003 at H_{loc}(1000 s), with a band that includes 0. The fitted H is lower in G than in its controls. The data do not exclude a gap of 0 for the straighter progress of group time once route, season, hour and wing are fixed, nor a gap as large as the upper end of the band.

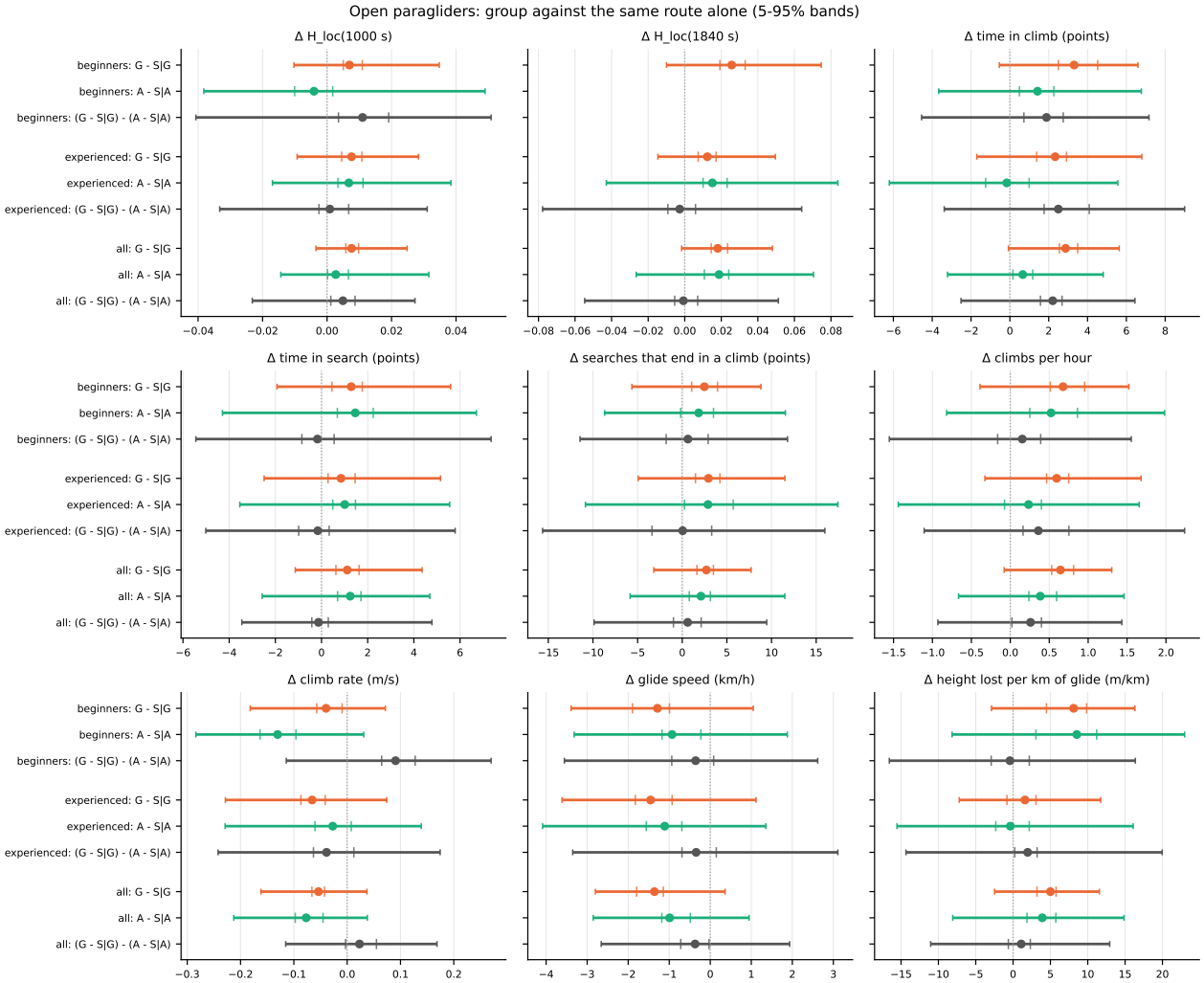


Figure 7: Same route, season, hour and wing: $\Delta(\mathrm{G}-\mathrm{S} \mid \mathrm{G})$, $\Delta(\mathrm{A}-\mathrm{S} \mid \mathrm{A})$ and their difference, by level of the wing (beginners, experienced, all). Dot: value; thick bar: 5–95% band of the draws of pilots × blocks; thin bar: range over the years, leaving one year out. $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and the seven statistics on dataset 1000 (phase statistics on the pieces with phases: 868 pieces of G at level all), $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on dataset 2250; average first, ratio after, unit the piece.

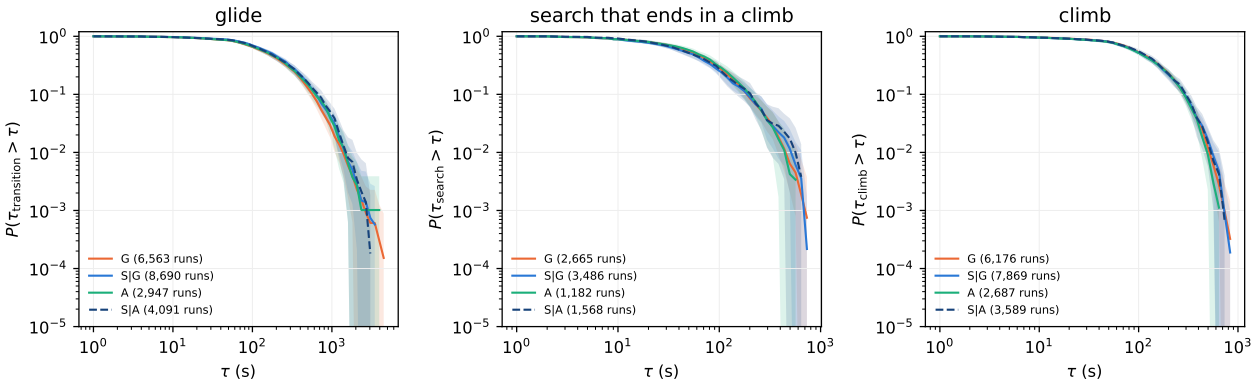


Figure 8: Same route, season, hour and wing, level all, dataset 1000: survival of glides, of searches ending in a climb and of climbs for G (runs: 6 176 climbs), its controls (7 869), A (2 687) and S | A (3 589). A run of a piece weighs 1, a run of a control 1/(controls of its piece); pointwise 5–95% bands.

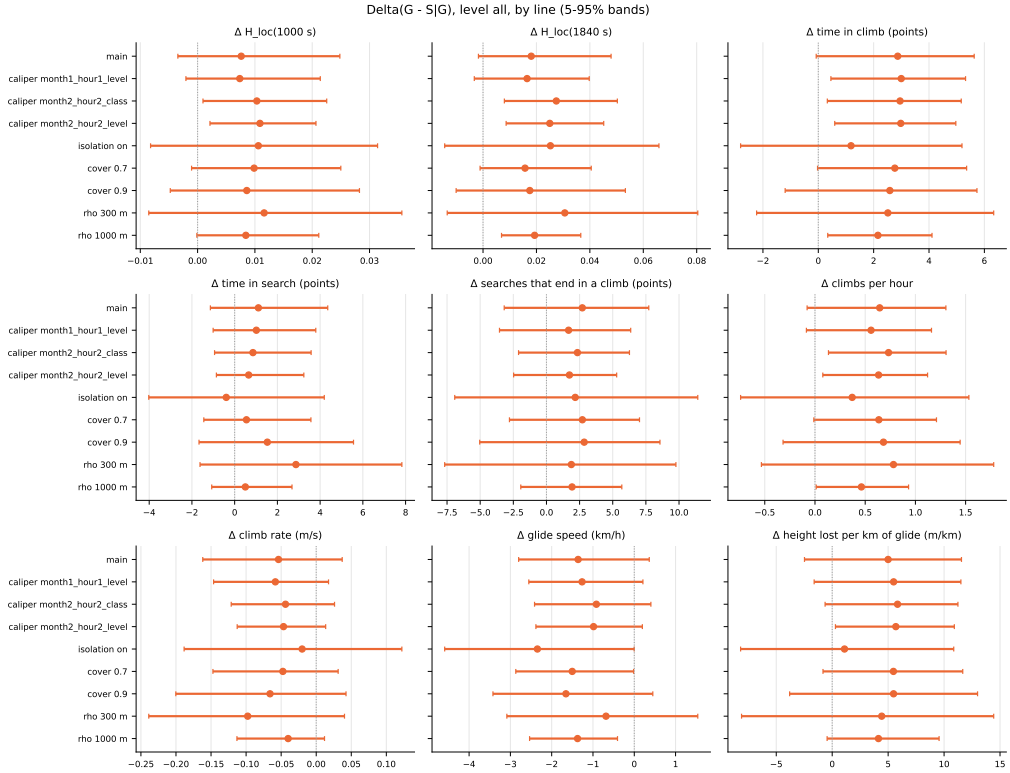


Figure 9: Robustness of $\Delta(\mathrm{G}-\mathrm{S} \mid \mathrm{G})$, level all, one row per line of the analysis: the other calipers of Table 6, isolation on (nobody within 10 km: 994 pieces), $\rho = 300$ and 1000 m, cover 0.7 and 0.9. The lines keep between 802 and 3 432 pieces on dataset 1000. 5–95% bands.

7 Point 3: how much each factor accounts for

How much of the raw gap between group time and solo flight do the route (R), the season and hour (T) and the wing (W) account for, and how much is left to the group?

Cells. For each of the 8 subsets C of $F = \{R, T, W\}$, $\Delta(C)$ is G minus the controls that satisfy only the conditions of C, on the same 1 763 pieces of G in every cell: those with a control in the cell F (the caliper of Section 6). $\Delta(\varnothing)$ compares the pieces with solo time of any route, season, hour and wing; $\Delta(F)$ is the result of Section 6. In the cells with R the controls are the matches that satisfy C. In the cells without R a control is a window of a solo piece of the pool, as long as the piece, drawn at random (at most 50 per piece) among the pool pieces that satisfy C and are long enough: with T, the pool piece in the month caliper and the window placed where its middle is within the hour caliper; with W, the same wing class (Appendix F). The Shapley value φ_k is the part of $\Delta(\varnothing)$ that comes from G and its controls differing in k (Appendix J):

$$\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W \qquad \text{(exact, on every draw).}$$

	G	raw gap	$\Delta(\varnothing)$		φ_R	φ_T	φ_W	$\Delta(F)$	
	pieces	point 1	controls	Δ				controls	Δ
H_{loc}(1000 s)	0.906	+0.018	0.908	−0.001	−0.006	+0.001	−0.004	0.899	+0.008
H_{loc}(1840 s)	0.893	+0.017	0.925	−0.032	−0.048	+0.001	−0.003	0.875	+0.018
time in climb (%)	29.2	+3.3	38.8	−9.5	−11.7	0.0	−0.7	26.3	+2.9
time in search (%)	15.3	−0.9	9.7	+5.6	+3.8	+0.1	+0.6	14.2	+1.1
searches ending in a climb (%)	49.5	+6.0	45.6	+3.9	+1.2	+0.8	−0.8	46.8	+2.7
climbs per hour	7.63	+0.72	6.94	+0.69	+0.01	+0.08	−0.05	6.98	+0.64
climb rate (m/s)	1.16	+0.044	1.09	+0.071	+0.095	+0.029	+0.001	1.22	−0.054
glide speed (km/h)	29.5	+2.1	35.2	−5.7	−3.6	+0.3	−1.0	30.9	−1.4
height lost per km of glide (m/km)	69	−18	95	−26	−34	0	+2	64	+5

Table 8: Decomposition, level all: value of G on its 1 763 pieces (868 for the phase statistics), the raw gap G − S of Section 5, the controls and Δ of the cells $\varnothing$ and F, and φ_R, φ_T, φ_W. Dataset 1000 (H_{loc}(1000 s), statistics), 2250 (H_{loc}(1840 s)); average first, ratio after, unit the piece. The eight cells in Appendix L.

The parts. For the time in climb, $\Delta(\varnothing)$ is −9.5 [−11.1, −8.0]: the pieces of G climb less than the controls of any route. The route accounts for −11.7 [−13.7, −9.7], season and hour for 0.0, the wing for −0.7, and $\Delta(F)$ is +2.9 [−0.1, +5.6]. The share of φ_R in $\Delta(\varnothing)$ is 122% and that of $\Delta(F)$ is −30%: they pass 100% and 0

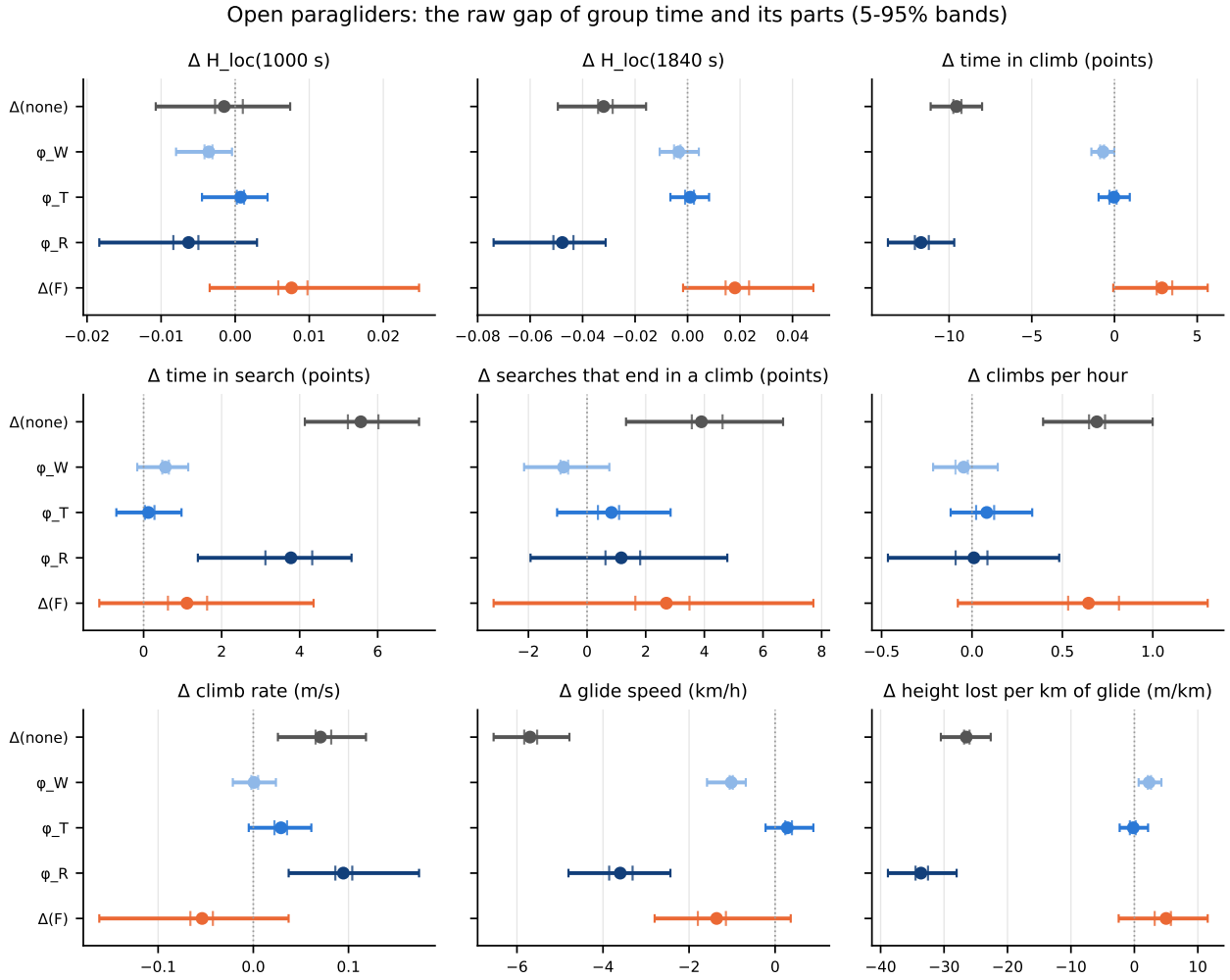


Figure 10: The raw gap and its parts, level all, on 1 763 pieces of G (868 with phase time for the seven statistics); $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000, $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on dataset 2250 (708 pieces). From the top: $\Delta(\varnothing)$, φ_W, φ_T, φ_R, $\Delta(F)$. Thick bar: 5–95% band of the same draws in every cell; thin: range over the years. Average first, ratio after, unit the piece.

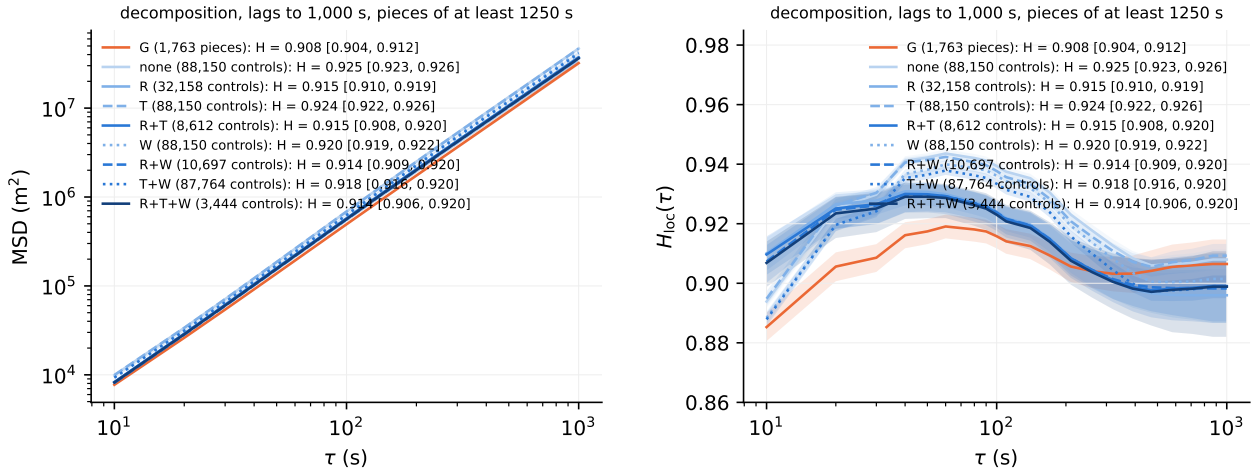


Figure 11: G (1 763 pieces) and the controls of the 8 cells (number of controls in the legend), dataset 1000 (pieces of at least 1 250 s): $\mathrm{MSD}(\tau)$ and $H_{\mathrm{loc}}(\tau)$, the darker the more conditions; 5–95% bands. Ensemble mean over pieces.

because $\Delta(F)$ has the opposite sign. Glide speed and height lost per km show the same picture of $\Delta(\emptyset)$ and φ_R ($\Delta(\emptyset)$ -5.7, φ_R -3.6, $\Delta(F)$ -1.4) and for the height lost per km of glide ($\Delta(\emptyset)$ -26, φ_R -34, $\Delta(F)$ $+5$). At $H_{\text{loc}}(1840\text{ s})$, $\Delta(\emptyset)$ is -0.032 [-0.049, -0.016] and φ_R is -0.048 [-0.074, -0.031]. At $H_{\text{loc}}(1000\text{ s})$, $\Delta(\emptyset)$ is -0.001 [-0.011, $+0.007$], near 0 with a band that includes it: the shares are not meaningful and Figure 10 shows the φ only; they are -0.006 (R), $+0.001$ (T) and -0.004 [-0.008, 0.000] (W).

Why $\Delta(\emptyset)$ is not the raw gap. The raw gap of Section 5 is $+3.3$ in time in climb, $+2.1$ km/h in glide speed, -18 m/km in height lost, $+0.018$ in $H_{\text{loc}}(1000\text{ s})$ and $+0.017$ in $H_{\text{loc}}(1840\text{ s})$; $\Delta(\emptyset)$ is -9.5, -5.7, -26, -0.001 and -0.032. Three things differ.

(a) *The pieces of* G. Only the pieces with a route match in the cell F enter, 1 763 of the 14 401 pieces of dataset 1000. Their $H_{\text{loc}}(1000\text{ s})$, time in climb and glide speed are lower than those of all of G: $H_{\text{loc}}(1000\text{ s})$ is 0.906 on them and 0.915 on all of G; time in climb is 29.2% against 36.3%, glide speed 29.5 against 35.2 km/h.

(b) *The controls without R.* Their windows are as long as the piece and come from solo pieces at least that long, so from long flights. These climb more and glide faster than the solo flights of Section 5: time in climb 38.8% in the controls of $\emptyset$ against 32.9% in S (34.2% with the ratio of sums, weighted by flight time), glide speed 35.2 against 33.2 km/h. $H_{\text{loc}}(1000\text{ s})$ of S rises with the length of the pieces, mostly beyond 2250 s: 0.897 on dataset 1000, 0.898 on 2250 and 0.912 on the longest; it is 0.908 in the controls of $\emptyset$.

(c) *The unit.* The unit is the piece, not the flight: the 1 763 pieces come from 1 648 flights, and the controls of a piece are averaged before the sum over pieces.

The controls with R have 26.9% of time in climb: a route fixes the flight from the start to the end of the piece, in the same direction, so φ_R carries that constraint together with the route.

What we keep. The raw gap does not carry through. Against controls of any route, group pieces climb less (-9.5) and glide slower (-5.7 km/h); the route accounts for the largest part of that (-11.7, -3.6 km/h), season and hour for almost none, the wing for a small part. What remains after the three factors, $\Delta(F)$, has a band that includes 0 for 9 of the 9 quantities of the figure, the time in climb at its edge ($+2.9$ [-0.1, $+5.6$]). At $H_{\text{loc}}(1000\text{ s})$ the raw gap is too close to 0 for shares to mean anything. The decomposition cannot attribute the raw gap to the group.

8 Point 4: before, during, after

Does the same flight move differently inside its group time than just before and after it?

Windows. For each stretch of group time of a flight, four windows of 2 400 s of the flight clock: *pre*, the 2 400 s before the stretch, if a solo stretch of the same flight at least that long precedes it; *start* and *end*, its first and last 2 400 s, if it lasts that long; *post*, the 2 400 s after it, if a solo stretch at least that long follows. A window is the unit. The MSD is measured on the 24 lags of the grid up to 1 840 s, the largest within $0.8 \times 2\,400$ s, with 56 origins at the longest. The windows of one stretch are one event (1 595 events with the four windows). The controls are two consecutive windows of 2 400 s drawn at random in the solo time of the flights of A (5 205 events) and in solo flights (26 175 events).

The MSD. In the windows inside the group $H_{\text{loc}}(1000\text{ s})$ is 0.923 (start) and 0.927 (end), against 0.902 before and 0.901 after (Figure 12). The entry difference $D_{\text{in}} = \text{start} - \text{pre}$ is $+0.018$ [$+0.013$, $+0.023$] and the exit difference $D_{\text{out}} = \text{end} - \text{post}$ is $+0.021$ [$+0.018$, $+0.025$], while after minus before is $+0.001$ [-0.005, $+0.007$]. The control pairs give -0.001 [-0.005, $+0.003$] (same flights alone) and $+0.001$ [0.000, $+0.003$] (solo flights).

The drift-free effect. If the quantity drifts linearly in time (the day advances, the pilot tires) the entry difference is the group effect plus the drift over one window, and the exit difference the group effect minus it. So

$$g = \frac{D_{\text{in}} + D_{\text{out}}}{2} \quad \text{(the group effect)}, \qquad d_w = \frac{D_{\text{in}} - D_{\text{out}}}{2} \quad \text{(the drift over one window)},$$

on the events that have the four windows (1 595 for the MSD, 1 150 with phase time; Appendix K). At $H_{\text{loc}}(1000\text{ s})$, g is $+0.023$ [$+0.018$, $+0.029$] and d_w is -0.002 [-0.006, $+0.002$].

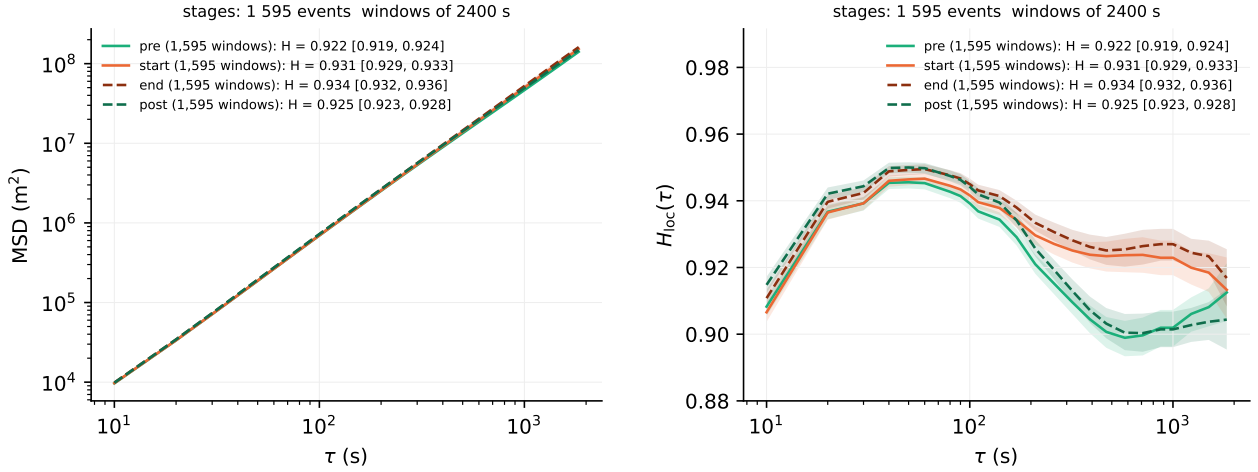


Figure 12: The four windows of the 1 595 events with the four windows (1 595 windows each), MSD(τ) (left) and $H_{\mathrm{loc}}(\tau)$ (right), lags to 1 840 s; ensemble mean over windows; 5–95% bands of the draws of dependency blocks. $H_{\mathrm{loc}}(1000\ \mathrm{s})$: pre 0.902, start 0.923, end 0.927, post 0.901.

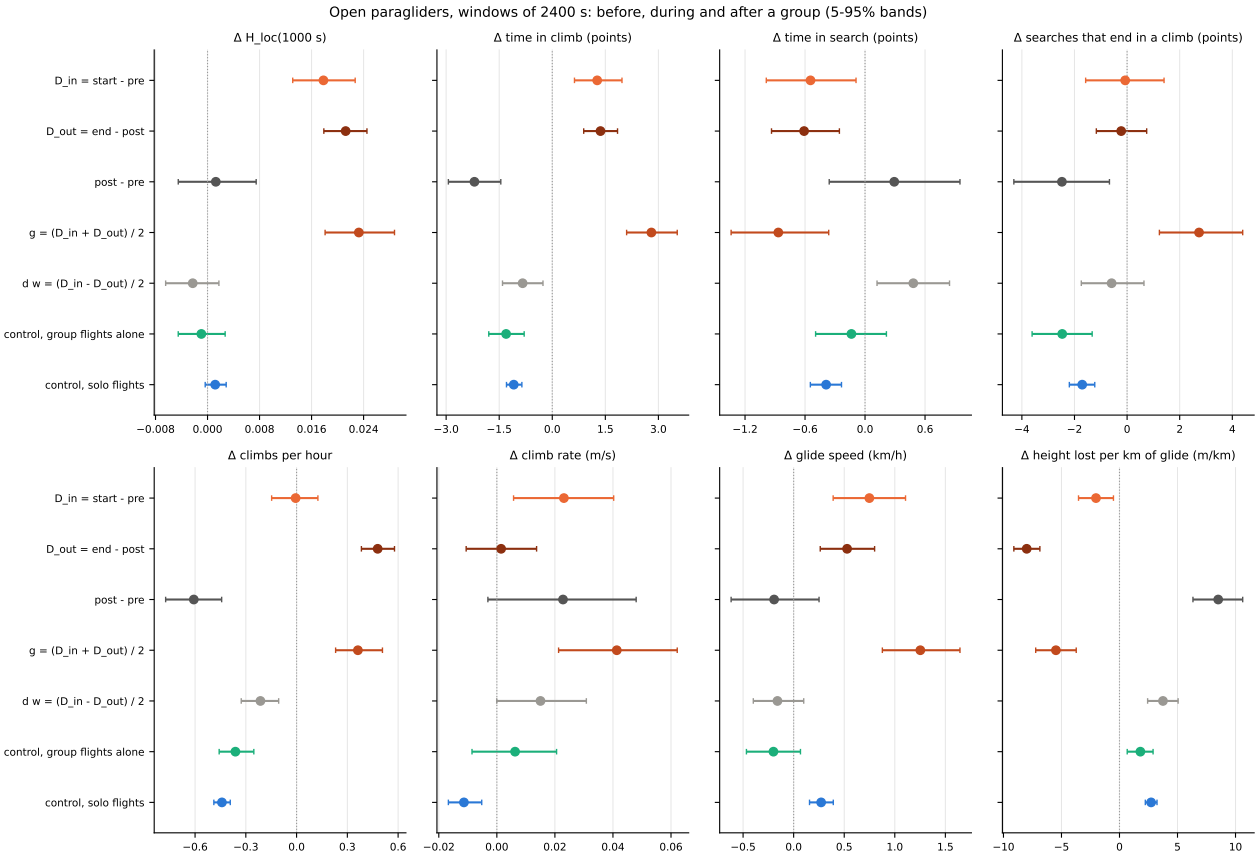


Figure 13: Before, during and after: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ and the seven statistics for D_{in}, D_{out}, post minus pre, g, d_w and the two control pairs; windows of 2 400 s, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on the MSD of the windows (1 595 events for g), statistics on 1 150 events for g (2 264 for D_{in}, 4 334 for D_{out}); average first, ratio after, unit the window; 5–95% bands of the draws of dependency blocks.

The phases. Time in climb rises on entry and on exit (D_{in} +1.3, D_{out} +1.4), and g is +2.8 [+2.1, +3.5] points; the control pairs give −1.1 (solo flights) and −1.3 (same flights alone). For the glide speed g is +1.3 [+0.9, +1.6] km/h and for the height lost per km of glide −5 [−7, −4] m/km; for the climbs per hour it is +0.36 [+0.23, +0.51].

	g, before and after	$\Delta(\text{G} - \text{S} \mid \text{G})$, same route	$g - \Delta$	same sign
$H_{\text{loc}}(1000\ \text{s})$	+0.023 [+0.018, +0.029]	+0.008 [−0.003, +0.025]	+0.016	yes
time in climb (%)	+2.8 [+2.1, +3.5]	+2.9 [−0.1, +5.6]	−0.1	yes
time in search (%)	−0.9 [−1.3, −0.4]	+1.1 [−1.1, +4.4]	−2.0	no
searches ending in a climb (%)	+2.7 [+1.2, +4.4]	+2.7 [−3.2, +7.7]	0.0	yes
climbs per hour	+0.36 [+0.23, +0.51]	+0.64 [−0.08, +1.30]	−0.28	yes
climb rate (m/s)	+0.041 [+0.021, +0.062]	−0.054 [−0.162, +0.037]	+0.095	no
glide speed (km/h)	+1.3 [+0.9, +1.6]	−1.4 [−2.8, +0.4]	+2.6	no
height lost per km of glide (m/km)	−5 [−7, −4]	+5 [−2, +12]	−10	no

Table 9: The drift-free effect g of this section (windows of 2 400 s, average first, unit the window) beside $\Delta(\text{G} - \text{S} \mid \text{G})$ of Section 6 (pieces, dataset 1000, same route, season, hour and wing, level all), at the same lag, with 5–95% bands; $g - \Delta$ and whether the two have the same sign.

Against the same route. At $H_{\text{loc}}(1000\ \text{s})$, g (+0.023) lies inside the band of $\Delta(\text{G} - \text{S} \mid \text{G})$ (+0.008 [−0.003, +0.025]), +0.016 above its point value. Time in climb, searches ending in a climb and climbs per hour have the same sign in the two comparisons (+2.8 against +2.9, +2.7 against +2.7, +0.36 against +0.64). Time in search (−0.9 against +1.1), climb rate (+0.041 against −0.054), glide speed (+1.3 against −1.4) and height lost per km of glide (−5 against +5) have opposite signs. The bands of g exclude 0 for those four and the bands of $\Delta(\text{G} - \text{S} \mid \text{G})$ include it (Table 9).

What we keep. Around the group the pilot's $H_{\text{loc}}(1000\ \text{s})$ rises on entry (+0.018) and on exit (+0.021) while the control pairs stay near 0, so g is +0.023 [+0.018, +0.029], with d_w −0.002 [−0.006, +0.002]. g lies in the band of the same-route $\Delta(\text{G} - \text{S} \mid \text{G})$. The two comparisons agree in sign for time in climb, searches ending in a climb and climbs per hour, and disagree for time in search, climb rate, glide speed and height lost per km of glide.

9 What we keep

	raw G − S	matched $\Delta(\text{G} - \text{S} \mid \text{G})$	selection $\Delta(\text{A} - \text{S} \mid \text{A})$	before and after g
$H_{\text{loc}}(1000\ \text{s})$	+0.018	+0.008 [−0.003, +0.025]°	+0.003 [−0.014, +0.032]°	+0.023 [+0.018, +0.029]
$H_{\text{loc}}(1840\ \text{s})$	+0.017	+0.018 [−0.002, +0.048]°	+0.019 [−0.027, +0.070]°	–
time in climb (%)	+3.3	+2.9 [−0.1, +5.6]°	+0.7 [−3.2, +4.8]°	+2.8 [+2.1, +3.5]
time in search (%)	−0.9	+1.1 [−1.1, +4.4]°	+1.2 [−2.6, +4.7]°	−0.9 [−1.3, −0.4]
searches ending in a climb (%)	+6.0	+2.7 [−3.2, +7.7]°	+2.1 [−5.8, +11.5]°	+2.7 [+1.2, +4.4]
climbs per hour	+0.72	+0.64 [−0.08, +1.30]°	+0.38 [−0.66, +1.46]°	+0.36 [+0.23, +0.51]
climb rate (m/s)	+0.044	−0.054 [−0.162, +0.037]°	−0.077 [−0.213, +0.038]°	+0.041 [+0.021, +0.062]
glide speed (km/h)	+2.1	−1.4 [−2.8, +0.4]°	−1.0 [−2.9, +0.9]°	+1.3 [+0.9, +1.6]
height lost per km of glide (m/km)	−18	+5 [−2, +12]°	+4 [−8, +15]°	−5 [−7, −4]

Table 10: The four comparisons side by side. Raw: G − S on the flights of Section 5 (dataset 1000 for $H_{\text{loc}}(1000\ \text{s})$, 2250 for $H_{\text{loc}}(1840\ \text{s})$, average first, unit the flight); same route, selection: pieces, level all, with 5–95% bands; before and after: g on windows. °: the band includes 0 (raw: the bands of G and S overlap). g has no value at $H_{\text{loc}}(1840\ \text{s})$: the windows are too short.

- Many flights declared open fold back on themselves, and the longest lags rest on those. An opening above 0.1 drops the ones that come back near their takeoff and keeps nearly the same share of lowland and mountain flights.
- **Raw.** On the open flights group time progresses straighter than solo flight: $H_{\text{loc}}(1000\ \text{s})$ 0.915 in G, 0.897 in S, 0.894 in A, and it has more time in climb (36.3% against 32.9%) and a faster glide (35.2 against 33.2 km/h). The bands of G and S do not overlap for $H_{\text{loc}}(1000\ \text{s})$, nor for 7 of the 7 statistics.
- **Same route, season, hour and wing.** The gap of $H_{\text{loc}}(1000\ \text{s})$ is +0.008 [−0.003, +0.025], of $H_{\text{loc}}(1840\ \text{s})$ +0.018 [−0.002, +0.048]. The bands include 0 for 9 of the 9 quantities, and those of the selection term $\Delta(\text{A} - \text{S} \mid \text{A})$ for 9 of them. The data do not separate group time from its controls on the same route.
- **Decomposition.** The route carries the largest part of $\Delta(\varnothing)$ for time in climb, glide speed and height lost per km of glide; season and hour carry almost none, and the wing a small part. $\Delta(\varnothing)$ differs from the raw gap because of the pieces, the controls without a route and the unit. At $H_{\text{loc}}(1000\ \text{s})$ it is near 0 and no share is given.

- **Before and after.** The drift-free effect g of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.023$ [$+0.018$, $+0.029$], inside the band of the same-route gap, $+0.016$ above its point value. Time in climb, searches ending in a climb and climbs per hour agree in sign with the same-route gap. Time in search, climb rate, glide speed and height lost per km of glide have the opposite sign; we give no reason for it.

Where a band includes 0 (Table 10, °) the comparison does not tell the sign.

A TA-MSD and ensemble MSD

For a flight f and a population, the pieces on the 10-s grid give the positions $\mathbf{r}_i$ (i the grid index, $\tau = k \cdot 10$ s). The time-averaged MSD pools the origins of all its pieces,

$$\overline{\delta_f^2}(k) = \frac{1}{N_f(k)} \sum_{\text{pieces}} \sum_i \left\| \mathbf{r}_{i+k} - \mathbf{r}_i \right\|^2,$$

$N_f(k)$ the number of pairs $(i, i+k)$ inside a piece. The ensemble MSD is the mean over the flights with a piece in the population, one weight per flight: $\mathrm{MSD}(\tau) = \frac{1}{F} \sum_f \overline{\delta_f^2}(\tau)$. The sums are computed with the fast Fourier transform and equal the direct sums.

Example. Positions on a line $\mathbf{r} = 0, 3, 4, 10$ (a single piece): at the lag of one step, MSD $= 15.33$, from the squared steps; at two steps, MSD $= 32.5$, from $(\mathbf{r}_2 - \mathbf{r}_0)^2$ and $(\mathbf{r}_3 - \mathbf{r}_1)^2$.

B Fitted H and H_{loc}

H is half the slope of $\log_{10}$ MSD against $\log_{10} \tau$, least squares with one weight per lag, over all the lags of a curve (10 s to the end of its dataset): $H = \frac{1}{2}$ slope. $H_{\mathrm{loc}}(\tau)$ is the same fit over the lags λ with $|\log_{10} \lambda - \log_{10} \tau| \le 0.25$; when fewer than 3 lags lie there, the 3 nearest. A straight line gives 1, a random walk $\frac{1}{2}$.

Example. With the two lags of one and two steps and the MSD above, the slope of the log–log line is $\log_2(32.5/15.33)$, so $H = 0.54$.

The window cut at the edge. $H_{\mathrm{loc}}(1840\ \mathrm{s})$ uses the lags within 0.25 decades of 1840 s, from 1 035 to $1840 \times 10^{0.25}$ s; dataset 2250 stops at 2250 s, so the window is one-sided and uses the lags from 1 035 to 2250 s only. Its band includes this truncation, since every draw is cut the same way. The window of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000 is cut in the same way at 1000 s.

C Datasets and the $\tau/0.8$ rule

A piece of duration ℓ gives, at lag τ, $(\ell - \tau)/10 + 1$ pairs of origins. Requiring $\ell \ge \tau/0.8$ gives origins over at least $0.2\,\ell$ of the piece, the same share of it at the longest lag the piece serves, whatever the piece. A curve to the lag T would mix populations if each lag kept the pieces long enough for itself (the long lags would rest on fewer, longer pieces). A dataset to T keeps the pieces of at least $T/0.8$ s and measures them at every lag up to T, so the same pieces carry the whole curve. Datasets 1000 and 2250 have $T = 1\,000$ and 2 250 s (pieces of at least 1 250 and 2 812.5 s); the longest has $T = 11\,450$ s (14 312.5 s), the largest lag of the grid at which each population keeps 100 flights. The windows of Section 8 have 2 400 s, so T is the largest lag of the grid up to $0.8 \times 2\,400$ s, 1 840 s.

D The phase statistics: average first against the ratio of sums

For a unit u (flight, piece, control segment or window) the terms n_u, d_u are the sums of the numerator and of the denominator over the runs whose middle lies in u, and T_u is the sum of the durations of its runs. The ratio of sums is $\sum_u n_u / \sum_u d_u$: a unit weighs by its d_u, hence by its phase time. The estimator used here is

$$\mathrm{stat} = \frac{\sum_u w_u\, n_u / T_u}{\sum_u w_u\, d_u / T_u}.$$

For the time shares and the climbs per hour the denominator d_u/T_u is the same constant for every unit, so the statistic is the mean over units of a share or a rate. For the others it is a ratio of means of rates per unit of phase time.

Example. Three units, searches that end in a climb out of all searches, $(n, d, T) = (1, 10, 100)$, $(5, 20, 400)$, $(2, 30, 500)$. The ratio of sums is $8/60 = 13.3\%$; the mean of the unit shares n/d is 13.9%; the estimator is $\sum n/T \,/\, \sum d/T = 12.6\%$.

	S	A	G
flights	27 665	6 397	7 896
phase time (h)	60 882	13 898	12 338
time in climb (%)	32.9 / 34.2	31.6 / 32.0	36.3 / 36.4
time in search (%)	11.4 / 10.3	11.6 / 10.3	10.5 / 9.9
searches ending in a climb (%)	40.5 / 43.1	42.1 / 44.8	46.6 / 47.2
climbs per hour	6.53 / 6.59	6.36 / 6.35	7.24 / 7.12
climb rate (m/s)	1.10 / 1.16	1.15 / 1.23	1.14 / 1.16
glide speed (km/h)	33.2 / 34.2	33.9 / 34.9	35.2 / 36.1
height lost per km of glide (m/km)	100 / 96	94 / 89	82 / 82

Table 11: Open flights, raw, pieces of dataset 1000, unit the flight: each statistic as *average first, ratio after / ratio of sums*, for S, A and G, with the flights and the phase time (h) they rest on.

E Survival function and its weighted version

For the durations x_j of the counted runs with weights w_j, $P(\tau_{\text{phase}} > t) = \sum_j w_j[x_j > t] \big/ \sum_j w_j$, on a logarithmic grid of t. The duration is the number of fixes of the run (1 Hz), trimmed at the 0.0001 and 99.9999 percentiles; a search counts only if the next run of the flight is a climb. In Section 5 every run weighs 1. In Section 6 a run of a piece weighs 1 and a run of a control segment weighs $1/m_g$, with m_g the number of controls of its piece g, so that the control side has the composition of the pieces. A draw of the bootstrap multiplies the weight of a run by the weight of its unit; the band is the pointwise 5–95% of the draws.

Example. Durations 10, 30, 30, 100 s with weight 1: $P(\tau > 20) = 0.75$ and $P(\tau > 30) = 0.25$.

F Route matching and calipers

A query piece (of G or A) is a path on the 10-s grid. A segment of a pool piece (an open flight with no contact of any length, on another date) matches it when

- *coverage*: at least 0.8 of the points of the query, one every 30 s, have a point of the pool track within $\rho = 500$ m;
- *direction and segment*: the points of the pool track nearest to the first and to the last point of the query are within ρ of them, and the first comes before the last: the pool flight flew the route the same way; the segment is the stretch between those points;
- the segment lasts at least as long as the dataset needs.

A query keeps one match per pool flight, the most covered. A caliper is then a filter on the match table: the month of the middles within m (circular), the local hour of the middles within h (circular), and the same wing class (A, B, C, D, CCC of the catalogue; a flight with no class is dropped) or the same level (beginners A and B, experienced C, D and CCC). The calipers are tried in the order of Table 6; the chosen one is the first with at least 200 pieces per level on dataset 2250. The robustness lines change one rule at a time: another caliper, isolation (at every point of the segment, no other flight within 10 km at the same minute), ρ of 300 or 1000 m, cover 0.7 or 0.9.

Cells without R (Section 7). The control is a window of a pool piece, with as many grid points as the piece of G, on another date. With T: the pool piece is within the month caliper and the window is placed at random among the positions whose middle is within the hour caliper (local time across midnight); without T: at random. With W: the pool piece has the wing class of the piece. At most 50 windows per piece, one per pool flight, drawn with a fixed seed. The same draws serve every cell.

G Reweighting to the piece

Piece g of G has terms a_g (its MSD sums, or its phase terms) and controls s with terms b_s. With w_g the weight of the piece (from the bootstrap) and w_s that of a control,

$$\text{group} = \sum_g w_g\, a_g, \qquad \text{control} = \sum_g w_g\, \frac{\sum_{s \in g} w_s\, b_s}{\sum_{s \in g} w_s}, \qquad \text{total} = \sum_g w_g,$$

over the pieces with positive weight and at least one control of positive weight. Each piece counts once on both sides, whatever the number of its controls: the control side has the composition of the pieces. The MSD at each lag is the mean group/total (resp. control/total), the logarithm and the slopes are taken after the means, and the phase statistics are the ratios $\sum_g w_g \bar{r}_g / \sum_g w_g \bar{q}_g$ with $\bar{r}_g$, $\bar{q}_g$ the means of r and q over the controls of the piece.

H Bootstrap of dependency blocks, and of pilots × blocks

Blocks. Flights that share an episode of the grouping are not independent, and neither are flights of the same site on the same day. Both tie flights into a block (a solo flight alone is a block). A draw takes B blocks with replacement (B the number of blocks, 1 000 draws, fixed seed) and gives each unit the count of its block as its weight. Each statistic and curve is recomputed on every draw, with the same draws for every quantity of a section; the band is the 5th and 95th percentile of the draws. *Pilots × blocks* (Sections 6 and 7). The pieces and the controls come from the same pilots, so two independent draws are made, of the pilots and of the blocks, and a flight weighs the product of the count of its pilot and the count of its block. A control flown by a pilot of the pieces is thus resampled with them.

I Leaving one year out

For each year of the archive the estimate is repeated with weight 0 on the flights of that year (the same form as a bootstrap weight). The thin bars of Figures 7 and 10 are the minimum and the maximum of these estimates: a result that depends on one year shows as a wide thin bar.

J The Shapley decomposition

Let $\Delta(C)$, $C \subseteq F = \{R, T, W\}$, be the gap of G against the controls that satisfy only the conditions of C. The part of the raw gap that comes from the factor k is

$$\varphi_k = \sum_{C \subseteq F \setminus \{k\}} \frac{|C|!\,(2-|C|)!}{3!} \left[\Delta(C) - \Delta(C \cup \{k\})\right],$$

with weights $\frac{1}{3}$ for $|C| = 0$, $\frac{1}{6}$ for $|C| = 1$ (two such sets) and $\frac{1}{3}$ for $|C| = 2$: they add to 1. They are the shares of the $3! = 6$ orders of the factors in which k comes after exactly the factors of C. Along one order $\Delta(\varnothing) - \Delta(F)$ is a sum of the three differences of consecutive cells, one per factor. Averaging over the six orders, $\sum_k \varphi_k = \Delta(\varnothing) - \Delta(F)$ exactly:

$$\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W.$$

Every order has the same weight, so the order in which the factors are added does not matter. The identity holds on every bootstrap draw (the maximum gap is 0 to the precision of the run), so the bands of the φ and of $\Delta(F)$ come from the same draws.

Additive example. If $\Delta(C) = g + \sum_{k \notin C} a_k$, then $\Delta(C) - \Delta(C \cup \{k\}) = a_k$ for every C, and $\varphi_k = a_k$. With $g = 0.01$, $a_R = 0.30$, $a_T = 0.05$, $a_W = 0.10$: $\Delta(F) = 0.01$, $\Delta(\varnothing) = 0.46$, and $\varphi_R = 0.30$, $\varphi_T = 0.05$, $\varphi_W = 0.10$. The shares $\varphi_k / \Delta(\varnothing)$ are meaningful only when $\Delta(\varnothing)$ is away from 0.

K The drift-free estimator

Let a quantity x drift linearly in time by δ per window, and let the group add γ inside the stretch. With four adjacent windows, pre then start at the entry, end then post at the exit,

$$D_{\text{in}} = x_{\text{start}} - x_{\text{pre}} = \gamma + \delta, \qquad D_{\text{out}} = x_{\text{end}} - x_{\text{post}} = \gamma - \delta,$$

because the end window comes before the post window. So $g = (D_{\text{in}} + D_{\text{out}})/2 = \gamma$ and $d_w = (D_{\text{in}} - D_{\text{out}})/2 = \delta$. The assumptions are a drift linear in time and a group effect that is the same on entry and on exit. Each of D_{in}, D_{out}, g and d_w is computed on the events that have the four windows, with the same bootstrap draws.

Example. $D_{\text{in}} = 3$ and $D_{\text{out}} = 1$ give $g = 2$ and $d_w = 1$.

L Figures of the other datasets and cells

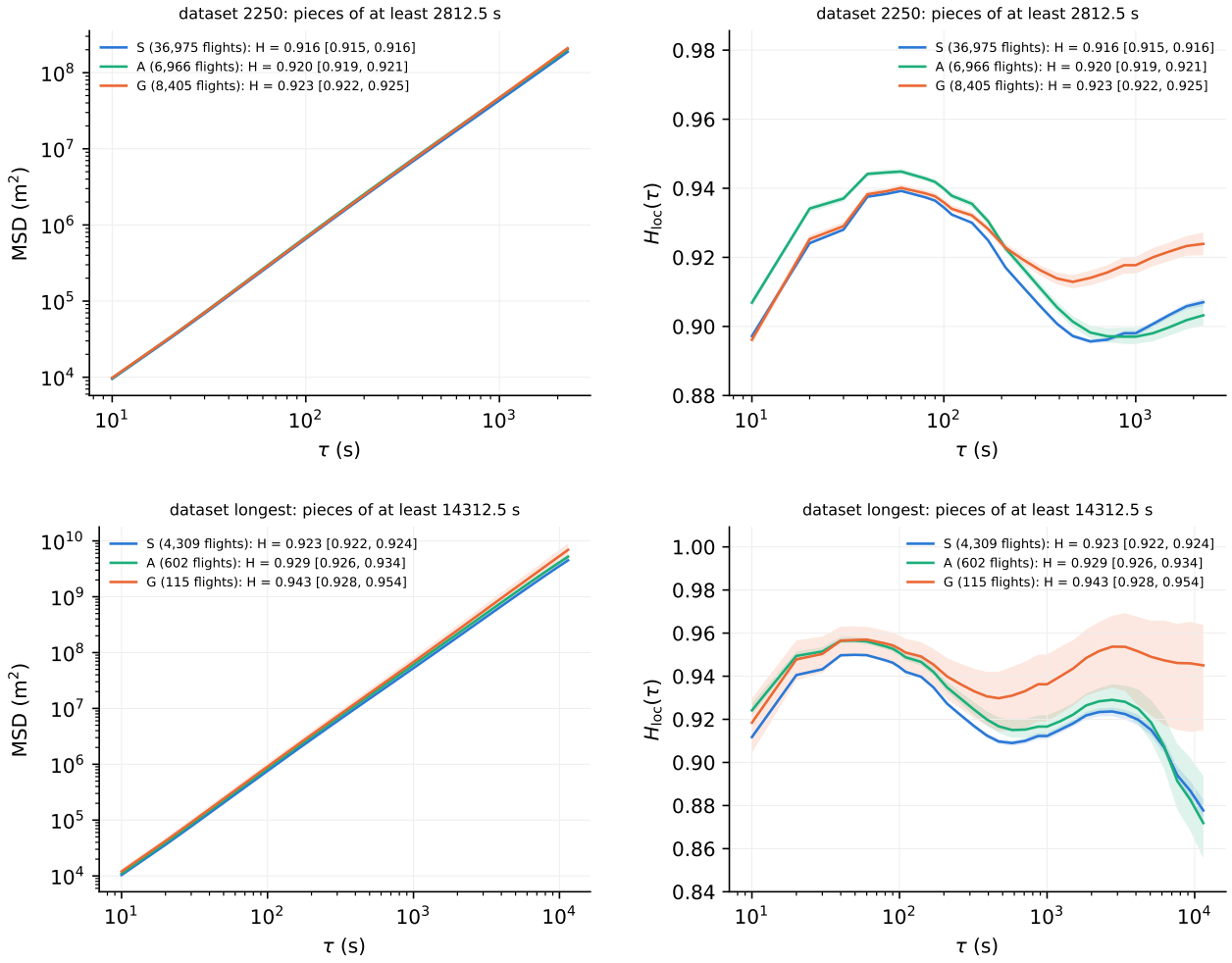


Figure 14: Raw, open flights, datasets 2250 (top: S 36 975 flights, A 6 966, G 8 405) and longest (bottom: 4 309, 602 and 115 flights): MSD(τ) and $H_{\text{loc}}(\tau)$, ensemble mean over flights.

	G	∅	R	T	RT	W	RW	TW	RTW
H_{loc}(1000 s)	0.906	0.908	0.896	0.909	0.899	0.901	0.898	0.899	0.899
H_{loc}(1840 s)	0.893	0.925	0.870	0.927	0.873	0.917	0.875	0.915	0.875
time in climb (%)	29.2	38.8	26.9	39.0	26.6	37.7	26.6	37.8	26.3
time in search (%)	15.3	9.7	13.6	9.7	13.9	10.5	14.0	10.6	14.2
searches ending in a climb (%)	49.5	45.6	47.3	47.3	46.9	44.5	46.5	45.8	46.8
climbs per hour	7.63	6.94	7.08	7.16	7.05	6.91	7.05	7.11	6.98
climb rate (m/s)	1.16	1.09	1.21	1.14	1.22	1.10	1.21	1.14	1.22
glide speed (km/h)	29.5	35.2	30.9	35.7	31.0	33.5	30.6	33.5	30.9
height lost per km of glide (m/km)	69	95	63	94	63	99	63	99	64

Table 12: Decomposition, level all, on 1 763 pieces of G (868 for the phase statistics): value of G and of the controls of each cell; the cells are named by the conditions they hold (∅ none, R route, T season and hour, W wing). Dataset 1000 for H_{loc}(1000 s) and the statistics, 2250 for H_{loc}(1840 s); average first, ratio after, unit the piece.

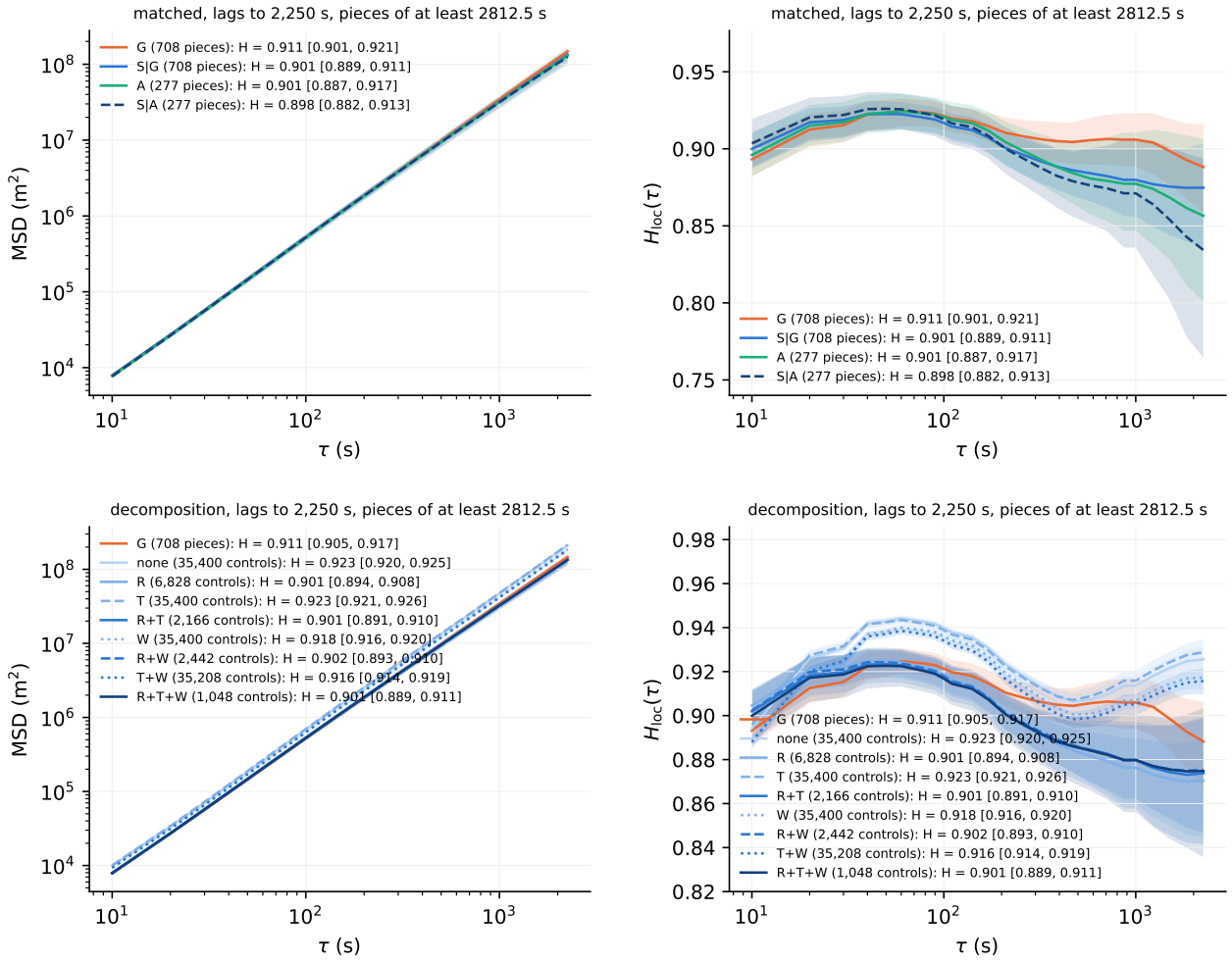


Figure 15: Top: same route, season, hour and wing, dataset 2250 (pieces of at least 2 812.5 s, 708 pieces of G). Bottom: G and the controls of the 8 cells, dataset 2250 (708 pieces). MSD(τ) and $H_{\text{loc}}(\tau)$, ensemble mean over pieces.

	∅	R	T	RT	W	RW	TW	RTW
H_{loc}(1000 s)	−0.001	+0.011*	−0.003	+0.007	+0.005	+0.008	+0.007	+0.008
H_{loc}(1840 s)	−0.032*	+0.023*	−0.035*	+0.020*	−0.024*	+0.018*	−0.022*	+0.018
time in climb (%)	−9.5*	+2.3*	−9.8*	+2.6*	−8.5*	+2.6*	−8.6*	+2.9
time in search (%)	+5.6*	+1.7	+5.6*	+1.4	+4.8*	+1.3	+4.7*	+1.1
searches ending in a climb (%)	+3.9*	+2.2	+2.2	+2.5	+5.0*	+3.0	+3.7*	+2.7
climbs per hour	+0.69*	+0.55*	+0.47*	+0.58	+0.72*	+0.58*	+0.52*	+0.64
climb rate (m/s)	+0.071*	−0.043	+0.023	−0.056	+0.067*	−0.044	+0.022	−0.054
glide speed (km/h)	−5.7*	−1.3*	−6.2*	−1.5*	−4.0*	−1.1	−4.0*	−1.4
height lost per km of glide (m/km)	−26*	+6*	−26*	+6	−31*	+6	−30*	+5

Table 13: Decomposition, level all: $\Delta(C) = \text{G} - \text{controls of the cell } C$ for the 8 cells (same pieces and units as Table 12); *: the 5–95% band excludes 0.

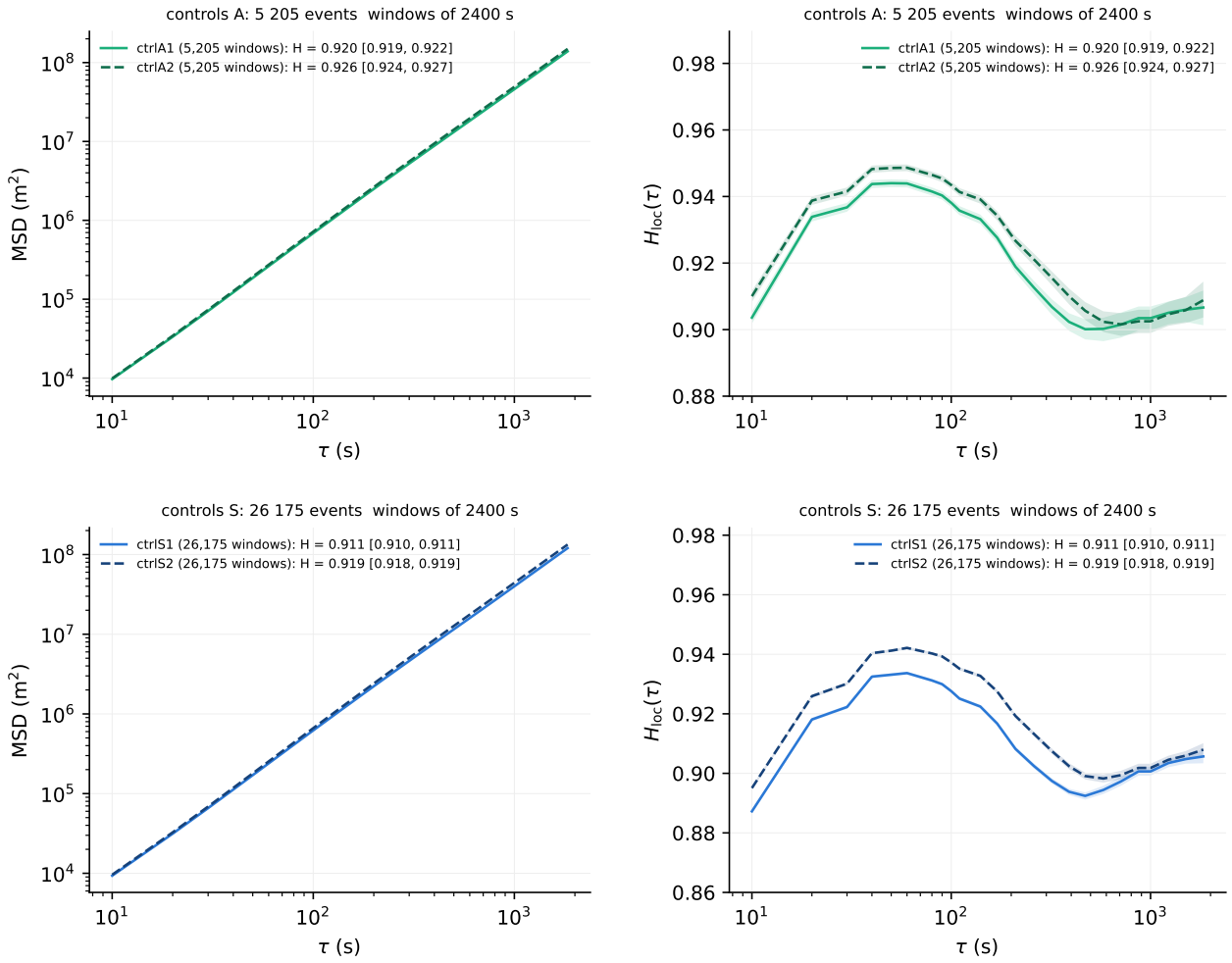


Figure 16: The control pairs of Section 8: two consecutive windows of 2 400 s in the solo time of the flights of A (top, 5 205 events) and in solo flights (bottom, 26 175 events); MSD(τ) and $H_{\mathrm{loc}}(\tau)$.